Emoji-Only Sentiment Analysis Using a Deep Learning Model: A Comparative Study with Text-Enhanced Approaches

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***Abstract*—**

Keywords—

I. INTRODUCTION

Sentiment analysis has emerged as a critical task in Natural Language Processing (NLP), enabling the automatic extraction of opinions, emotions, and attitudes from large-scale textual data. With the exponential growth of social media platforms, user-generated content has evolved into a highly informal and multimodal form of communication. In this context, emojis have become an integral component of digital expression, serving as compact yet powerful carriers of affective meaning that complement or even replace textual cues.

Conventional sentiment analysis approaches have predominantly relied on text-based representations, employing machine learning and deep learning models such as Long Short-Term Memory (LSTM) networks and transformer-based architectures like BERT. These models have demonstrated strong capabilities in capturing contextual semantics and sequential dependencies [1], [2]. However, they often fail to fully represent the nuances of modern communication, where emojis significantly influence sentiment interpretation. Emojis can reinforce, alter, or even invert the polarity of textual content, particularly in informal or sarcastic contexts, leading to potential misclassification when ignored or underrepresented.

To address these limitations, recent studies have explored fusion-based approaches that integrate textual and emoji information. While these methods improve classification performance by leveraging additional emotional cues, they inherently treat emojis as auxiliary features rather than primary sentiment carriers. Furthermore, existing research on emoji-only sentiment analysis remains limited and often constrained to domain-specific datasets or traditional modeling techniques. Challenges such as context-dependent emoji semantics, sarcasm interpretation, and the lack of comprehensive evaluation frameworks continue to hinder progress in this area [3], [4].

In response to these challenges, this study proposes a unified sentiment analysis framework that systematically evaluates three distinct modalities: text-only, emoji-only, and combined text–emoji inputs. The proposed approach employs hybrid BERT-LSTM architecture to effectively

capture both contextual representations and sequential dependencies. In addition, a dataset is used, specifically designed to include emoji-dense content and sarcasm, enabling a more realistic and challenging evaluation of sentiment models in contemporary communication settings.

The main contributions of this paper are as follows. First, a unified experimental framework is developed to enable a fair comparison between text-only, emoji-only, and fusion-based sentiment analysis approaches. Second, the study investigates the viability of emoji-only sentiment analysis as an independent modality, addressing a significant gap in existing literature. Third, a sarcasm-focused and emoji-rich dataset is constructed to evaluate model robustness under complex linguistic conditions. Finally, a hybrid BERT-LSTM architecture is utilized to enhance the modeling of contextual and semantic relationships within multimodal data.

The remainder of this paper is organized as follows. Section II presents the related work. Section III describes the proposed methodology. Section IV details the experimental setup and results. Section V discusses the findings, and Section VI concludes the paper with directions for future research.

II. LITERATURE REVIEW

A. Text-Based Sentiment Analysis

Text-based sentiment analysis is the traditional approach for analyzing opinions in online content. Early methods relied on machine learning algorithms such as Naive Bayes, Support Vector Machines, and Logistic Regression with bag-of-words and TF-IDF features. With advancements in deep learning, LSTM and transformer-based models such as BERT have become dominant due to their ability to capture contextual and sequential dependencies.

BERT demonstrates superior performance in understanding contextual semantics and complex linguistic patterns [1], while Bi-LSTM effectively models sequential dependencies across domains such as social media and software engineering [2].

However, text-based methods fail to fully capture modern communication, where emojis play a crucial role. Ignoring emojis leads to loss of emotional information and reduced performance [1]. Studies further show that incorporating emoji features improves classification accuracy, indicating that text-only models overlook important sentiment cues [5].

Thus, text-based approaches remain limited in emoji-rich environments, motivating the need for emoji-enhanced and emoji-only methods.

B. Fusion-Based Sentiment Analysis (Text + Emoji)

Fusion-based approaches integrate text and emojis to overcome limitations of text-only models, recognizing emojis as key carriers of emotional context. Studies show that combining both modalities improves sentiment classification by capturing richer user intent [6].

Hybrid ML/DL models demonstrate that emoticons enhance performance and can dominate sentiment interpretation [7], while multi-view frameworks treat text and emojis as complementary inputs [8]. Advanced architectures such as EMFSA (cross-attention) and multimodal systems combining BERT, Bi-LSTM, and ResNet further improve accuracy, with performance dropping when emoji features are removed [9], [10].

Additional studies confirm that integrating emojis improves performance across ML and DL models, with transformer-based approaches achieving the best results [11]. Fusion methods are also effective in handling sarcasm and nuanced expressions [12].

However, these approaches rely primarily on text and treat emojis as supplementary features, without exploring their standalone capability.

C. Emoji-Based Sentiment Analysis

Recent research shows that emojis are semantically rich and capable of independently conveying sentiment. Large-scale studies demonstrate that emoji-only models can achieve performance comparable to text-based models, with equal accuracy in Logistic Regression (0.75) and slightly lower performance in BiLSTM (0.74 vs. 0.80), while requiring up to 40× less data and faster processing [4].

Deep learning studies confirm strong correlations between emojis and emotional states, with models like BERT effectively learning text–emoji relationships. However, challenges such as class imbalance, sarcasm, and cultural variation remain [3].

Other works report emoji-only accuracy around 70–85%, with limitations in handling semantic relationships and unseen emojis [13]. Nevertheless, emojis can significantly influence sentiment, even inverting polarity in up to 40% of cases [14].

Incorporating emojis improves performance across models (e.g., BERT accuracy from 95.3% to 96.2%) [11], while traditional ML approaches show improvements up to

26.7% [15], [16]. Rule-based systems like VADER struggle with sarcasm, and frameworks such as SEntiMoji show that emojis can act as weak supervision signals for deep learning [2].

Overall, emojis are powerful sentiment indicators capable of functioning independently and enhancing sentiment understanding.

D. Research Gap

Despite progress, several key gaps remain.

First, most studies focus on fusion approaches, treating emojis as supplementary, with limited exploration of emoji-only analysis and reliance on domain-specific datasets [4], [11], [17].

Second, emoji-only models are mainly based on traditional ML or simple DL, and even advanced models struggle with semantic relationships and unseen emojis, highlighting the need for architectures such as BERT-LSTM [1], [13].

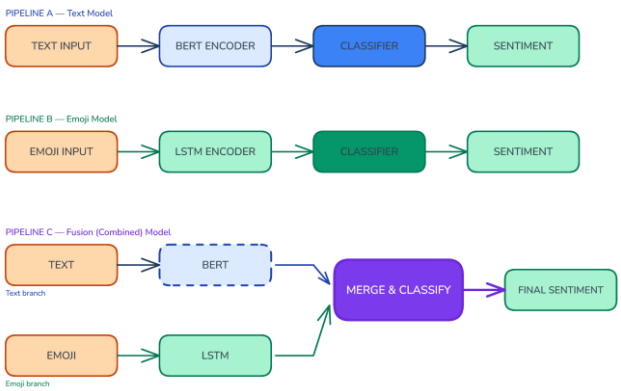
Third, sarcasm and irony are insufficiently addressed, as few studies use datasets specifically designed for emoji-only sarcasm detection [3], [18].

Fourth, many approaches rely on static emoji representations, limiting context-aware and culturally adaptive understanding [15], [16].

Finally, there is no unified framework comparing text-only, emoji-only, and fusion approaches under the same conditions. No prior work integrates all three pipelines within a single BERT-LSTM framework evaluated on sarcasm-heavy, emoji-dense datasets.

III. METHODOLOGY

This study employed a unified experimental framework to evaluate deep learning models for sentiment classification across three input modalities: text-only, emoji-only, and a hybrid approach. Three pipelines were developed for comparative analysis.



- **Pipeline A (Text-Only):** A pre-trained BERT model was used as a fixed feature extractor with frozen layers. A custom Multi-Layer Perceptron (MLP) classifier was trained on the CLS token representation for sentiment prediction.
- **Pipeline B (Emoji-Only):** A custom model processed emoji sequences using an Embedding layer followed by a Bidirectional Long Short-Term Memory (Bi-LSTM) network.
- **Pipeline C (Combined):** A dual-branch architecture integrated BERT-based text features and Bi-LSTM-based emoji features. The concatenated representations were passed through a fusion layer to generate final prediction.

Data Preparation

Three datasets were used:
Primary dataset (100,000 samples) for training/testing
Two external validation sets:
External Dataset 1 (2,000 samples)
External Dataset 2 (40,000 samples).

Preprocessing included regex-based text cleaning and emoji extraction using the emoji Python library. Emojis were converted into textual aliases (e.g., :smiling_face:). Sentiment labels were mapped to three classes: Negative (0), Neutral (1), and Positive (2).